Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Mid2 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** The simplest form of ((-3y)^2) / ((-y)^5) is:
A. 9 **B.** -9 **C.** -9y^-3 **D.** 9 / (y^3)

____ **2.** If log_3(y^2) = 2, then what is the value of y?
A. 9 **B.** -9 **C.** 3 **D.** 0

____ **3.** If 4^x = 1/16, then what is x?
A. 1 **B.** -1 **C.** -2 **D.** 2

____ **4.** The simplest form of 5th-root(243) is:
A. 1/3 **B.** 3 **C.** 5 **D.** 1/5

____ **5.** If (9^x)(4^x) = 6*sqrt(6), then x =
A. 3 **B.** 3/4 **C.** 4/3 **D.** 4

____ **6.** For any positive real number x, which one is equal to (log_a x)^2?
A. 2 log_a x **B.** log_a (x^2)
C. (log_a x) / 2 **D.** log_a x * log_a x

____ **7.** For what value of x is the equation 3^(x-1) = sqrt(3)?
A. 2 **B.** 1/2 **C.** 2/3 **D.** 3/2

____ **8.** If log_a x = log_x a for positive numbers x and a (x, a ≠ 1), then x =
A. a^2 **B.** sqrt(a) **C.** { } **D.** 1/a

____ **9.** If 3*(9/25)^(x-1) = 5*(27/125)^x, then x =
A. 0 **B.** 1 **C.** 2 **D.** -1

____ **10.** What is the value of x in log_2 x + log_2(x + 2) = log_2 8?
A. -4 **B.** 2 **C.** 6 **D.** A & B

____ **11.** The simplest form of (3/8) * (log_(1/8) (1/128)) is:
A. 7/8 **B.** 3/8 **C.** 7/3 **D.** 64/63

____ **12.** If log_4 x - log_4(x - 1) = log_2 8, then find x:
A. 64/63 **B.** 63/64 **C.** 8 **D.** 4

____ **13.** If log_4 x = 1/2, then what is the value of x?
A. 2 **B.** 16 **C.** 1/2 **D.** 4

____ **14.** What is the simplest form of ((3rd-root(32))^2) / (4^(5/3))?
A. 1 **B.** 2 **C.** 4 **D.** 1/2

____ **15.** If log 2 = 0.3010 and log 3 = 0.4771, then find log_(1/3) (0.0002):
A. 7.753 **B.** 5.658 **C.** -7.753 **D.** 2.699

____ **16.** Workout: What is the simplest form of log_4(sqrt(16))?
A. 1 **B.** 2 **C.** 1/2 **D.** 4

____ **17.** Workout: Simplify ((n^2) * (n^3)) / (n^-2):
A. n^7 **B.** n^3 **C.** n^-1 **D.** n^6

____ **18.** Workout: If log_2(x - 1) + log_2 3 = 3, then find x:
A. 11/3 **B.** 3/11 **C.** 5 **D.** 7/3

___ **19.** Workout: Evaluate log 4 + log 25:

A. 2 **B.** 100 **C.** 1 **D.** 10

___ **20.** Workout: If log_x 3 = 1/3, then find x:

A. 27 **B.** 9 **C.** 1 **D.** 81